Notes of Model Identification and Data Analysis

A series of notes on the course MIDA modulo 1 as taught in Politecnico di Milano by Sergio Bittanti during the academic year 2018/2019

Notes of Model Identification and Data Analysis
Side Notes
- Childhood Memories
- Observations
- Doubts
- Canonical Form

Exercises on Covariance & Spectra
- Covariance
  - General method to compute variance and covariance of a process
  - Draw the diagram of the prediction error variance as a function of the prediction horizon k
- Spectra
  - Compute the spectrum via definition
  - Compute the spectrum via magic formula

Exercise on Minimum Variance Controller
Optimal Estimates of Parameters via PEM
  - Tips
  - For MA Processes
  - For AR Processes

Parameter Estimation
- Knowing Variance
- Knowing Measurements

Optimal Predictors via Long Division (only $e(t)$ case)

Side Notes

Childhood Memories

- $ae^{jw} = acos(w) + jasin(w)$
- $|z_1|^2 = z_1 \bar{z}_1$
- $e^{j\omega} + e^{j(-\omega)} = 2cos(\omega)$
- ***Gain of a transfer function***
  if it's mathematically doable, $\mu = G(s=0) = G(jw=0)$, so in model analysis $\mu = G(z=1)$ since $z = e^{jw}$
- ***From the algebraic form to the exponential one***
  $z = a + jb = re^{jw}$

$$r = \sqrt{a^2 + b^2}$$

$$\theta \in (-\pi, \pi] = \begin{cases} \frac{\pi}{2} & se\ a = 0, b > 0 \\ -\frac{\pi}{2} & se\ a = 0, b < 0 \\ non\ definito & se\ a = 0, b = 0 \\ \arctan(\frac{b}{a}) & se\ a > 0, b \geq 0 \\ \arctan(\frac{b}{a}) + \pi & se\ a < 0, b \geq 0 \\ \arctan(\frac{b}{a}) - \pi & se\ a < 0, b < 0 \end{cases}$$

$$\theta \in [0, 2\pi) = \begin{cases} \frac{\pi}{2} & se\ a = 0, b > 0 \\ \frac{3\pi}{2} & se\ a = 0, b < 0 \\ non\ definito & se\ a = 0, b = 0 \\ \arctan(\frac{b}{a}) & se\ a > 0, b \geq 0 \\ \arctan(\frac{b}{a}) + 2\pi & se\ a > 0, b < 0 \\ \arctan(\frac{b}{a}) + \pi & se\ a < 0, b\ qualsiasi \end{cases}$$

- ***Frequency Response Theorem***
  Let $u(t) = A\sin(\bar{w}t + \varphi)$. If the system is asymptotically stable, at the end of the initial transitory, the output is sinusoidal as well.

$$y(t) = B\sin(\bar{w}t + \theta)$$

$$\begin{cases} B = A|G(j\bar{\omega})| \\ \theta = \varphi + \angle G j\bar{\omega} \end{cases}$$

Observations

- in an $MA(n)$ all the covariances $\gamma(\tau)$ with $\tau > n$ are equal to zero: $\gamma(tau > n) = 0$
- A process is stationary if its transfer function is stable (poles and zeros $\leq |1|$) and it's fed by a stationary signal (i.e., a white noise).
  NB: if the transfer function is in canonical form, it is surely stable.
- $2\eta(t)\ \ \eta \sim WN(0, \lambda^2) \rightarrow \tilde{\eta} \sim WN(0, \lambda^2 2^2)$

- In case the transfer function is fed by a white noise with $E[\eta(t)] = 0$ we can simplify the expression in: $\gamma(y(t)) = E[\ (\ y(t)\ )^2\ ]$

Doubts

- is c_0 in MA processes always equal to 1?
- how to find out if a closed-loop system is stable?
- why the results of the predictor found via PEM or via long division are different? when should I use one and why should I use the other? (page 28 exams pdf)
- if a signal has a zero, does its spectrum surely goes to zero at certain frequencies ? (page 33 exams pdf)

Canonical Form

A stochastic process is in canonical form if its transfer function has the following properties:

- $N(z)$ and $D(z)$ have the same order
- $N(z)$ and $D(z)$ are monic (the coefficient of the highest degree variable is 1)
- $N(z)$ and $D(z)$ are coprime (no roots in common)
- zeros and poles are inside the unitary circle ($|z| \leq 1$ and $|p| \leq 1$)

Exercises on Covariance & Spectra

Variance and Spectra Properties

If two signals α and β are independent and $\varphi = \alpha + \gamma$:

- $\gamma_\varphi = \gamma_\alpha + \gamma_\beta$
- $\Gamma(\varphi(t)) = \Gamma(\alpha(t)) + \Gamma(\beta(t))$

All Pass Filter

$y(t) = \frac{1+az}{1+\frac{1}{a}z}\varphi(t) = (\frac{1}{a}\frac{1+az}{1+\frac{1}{a}z})a\varphi(t) = a\varphi(t)$

Covariance

General method to compute variance and covariance of a process

Transform your process in time domain:
$y(t) = a_1 y(t-1) + a_2 y(t-2) + c_o\eta(t) + c_1\eta(t-1)$

If we recognize that it is an $MA(n)$, we can use a shortcut. We wrote it somewhere, go find it.

Compute $E\big[y(t)\big]$. if it is different from zero, transform it into an equivalent process with expected value equal to zero. {TODO}

Since $E\big[y(t)\big] = 0$ we see that
$\gamma_y(0) = E\big[y(t)^2\big] - E\big[y(t)\big]^2 = E\big[y(t)^2\big]$

$$\gamma_y(k) = E\Bigg[\Big(y(t) - E[y(t)]\Big)\Big(y(t-k) - E[y(t-k)]\Big)\Bigg] = E\Big[y(t)y(t-k)\Big]$$

In order to compute the variance of y do the following:

$$\gamma_y(0) = E[y(t)y(t)] = E\Bigg[y(t)\Big(a_1 y(t-1) + a_2 y(t-2) + c_o\eta(t) + c_1\eta(t-1)\Big)\Bigg]$$

now multiply all the members with one another and take in mind that:

- $E\Big[y(t-k)\eta(t-j)\Big] = 0$ if $k > j$
- $E\Big[y(t-k)\eta(t-k)\Big] = E\Big[y(t)\eta(t)\Big] = E\Bigg[\Big(a_1 y(t-1) + a_2 y(t-2) + c_o\eta(t) + c_1\eta(t-1)\Big)\eta(t)\Bigg]$
- Once you compute $\gamma_y(0)$ or $\gamma_y(1)$ (or even other covariances...) you could obtain values dependent from other covariances, for example $\gamma_y(0) = 2\gamma_y(1)$, but don't worry, in the end you'll have a system of n unknown covariances and n equations.

Compute the Variance of an AR(1) process $y(z) = \frac{1}{A(z)}\eta(z)$

1. Transform the function in the domain of time

2. knowing that $\gamma_{y(t)}(0) = \gamma_{y(t-1)}(0)$ and that $y(t) = y(t-1) + \eta(t)$ then $\gamma_y(0) = \gamma_y(0) + \gamma_\eta(0)$ if and only if y and η are independent, we can solve the last equation because usually the variance of η is known. Another important property of the variance is that $\gamma(ay(t)) = a^2\gamma(y(t))$, given that a is a constant.

Draw the diagram of the prediction error variance as a function of the prediction horizon k

If they ask you such question, probably they already asked you to do the long division for k steps.

Well, use the information gathered from that task, you'll use $Q_1, Q_2, \ldots, Q_k$.

keep in mind that

$$\varepsilon(t|t-k) = Q_k(z)\tilde{e}(t)$$

where $\tilde{e}(t)$ is the WN of the process in canonical form.

Q_k are all known, so just propagate the variance within the equation: $Var(\varepsilon(t|t-k)) = Var(Q_k\tilde{e}(t))$

To do the diagram just put the k in the x axis, and $variance(\varepsilon)$ in the y axis.

Spectra

Compute the spectrum via definition

$$\Gamma(\omega) = \sum_{t=-\tau}^{+\tau} \gamma(t)e^{-j\omega t}$$

Compute the spectrum via magic formula

$$\Gamma(\omega) = |W(e^{jw})|^2 var(\eta(t))$$

Exercise on Minimum Variance Controller

(This is probably just for MIDA2)

First step: compute the k-steps predictor

1. Make sure to be in canonical form.
2. Compute the long division of the transfer function of η, which is $\frac{C(z)}{A(z)}$. we stop at the k-th iteration and we obtain R_k and Q_k (sometimes called E_k).
3. the k-steps predictor is:
   $\hat{y}(t|t-k) = \frac{\hat{R}_k}{C(z)} y(t-k) + \frac{B(z)Q_k}{C(z)} u(t-k)$

Knowing the predictor $y(t+2|t)$, design the Minimum Variance Controller for the system, considering a reference signal $y^0(t)$.

$$y(t) = \frac{C(z)}{A(z)}\eta(t) + \frac{B(z)}{A(z)}u(t-k)$$

We set $\hat{y}(t+2|t) = y^0(t)$.
We are able to set it this way if and only if $B(z)$ has $|z| < 1$.

Isolate $u(t)$.

Draw the block diagram of the control system X
the input is $y^0(t)$, the output is $y(t)$. take the expression of y that is provided to you by the text.

Optimal Estimates of Parameters via PEM

Tips

- sometimes, in order to solve the systems it's handy to compute, for example, $(c_1 + c_2)^2 = \frac{\gamma(0)+2\gamma(1)}{var(e(t))}$ and $(c_1 - c_2)^2 = \frac{\gamma(0)-2\gamma(1)}{var(e(t))}$.

For MA Processes

Suppose to have infinitely many data generated by the following system:

$$\delta : y(t) = 2\eta(t) + 4\eta(n-2)$$

where $\eta(t) \sim WN(0.1)$

For such data, consider the model classes:

$$M_1(a) : y(t) = a_1 y(t-1) + \xi_1(t)$$

$$M_2(a) : y(t) = a_1 y(t-2) + \xi_2(t)$$

where $\xi_1(t) \sim WN(0, \lambda_1^2)\ and\ \xi_2(t) \sim wn(0, \lambda_2^2)$

By using the prediction error minimization method, find the optimal estimates of parameters a_1 and a_2.

in this case we have a $MA(2)$ for δ.

This means that we can compute the covariances in the following way:
Knowing that $y(t) = c_0\eta(t) + c_1\eta(t-1) + \ldots$

$$\gamma(0) = (c_0^2 + c_1^2 + \ldots c_{n-1}^2)\gamma(\varepsilon(t))$$

$$\gamma(1) = (c_0c_1 + c_1c_2 + \ldots + c_{n-2}c_{n-1})\gamma(\varepsilon(t))$$

$$\gamma(2) = (c_0c_2 + c_1c_3 + \ldots + c_{n-3}c_{n-1})\gamma(\varepsilon(t))$$

$$\gamma(3) = (c_0c_3 + c_1c_4 + \ldots + c_{n-4}c_{n-1})\gamma(\varepsilon(t))$$

Let's consider only the first model:

$$M_1(a) : y(t) = a_1 y(t-1) + \xi_1(t)$$

we want it in the form:

$$y(t) = \frac{C(z)}{A(z)}\xi_1(t)$$

and we obtain:

$$C(z) = 1; \quad A(z) = 1 - a_1 z^{-1}$$

we can now write the predictor of y:

$$\hat{y}(t|t-1) = \frac{C(z) - A(z)}{C(z)} y(t) = a_1 z^{-1} y(t)$$

And thus compute the prediction error ε:

$$\varepsilon(t) = y(t) - \hat{y}(t|t-1) = y(t) - a_1 z^{-1} y(t)$$

Now we can compute J, the performance index based on the prediction error:

$$J = E[(\varepsilon(t))^2] = E[(y(t) - a_1 y(t-1))^2] = E[y^2(t)] + a_1 E[y^2(t-1)] - 2a_1 E[y(t)y(t-1)]$$

We know that, being $\eta \sim WN(0,1)$, we have $E[y^2(t)] = \gamma_y(0) = E[y(t-1)]$

so:

$$J = \gamma_y(0) + a_1^2 \gamma_y(0) - 2a_1 \gamma_y(1)$$

and we derive J wrt a_1 and equal it to zero

$$\frac{\delta J}{\delta a_1} = \ldots = 2(-\gamma_y(1) + a_1 \gamma_y(0)) = 0$$

Thus obtaining

$$\hat{a}_1 = 0$$

We repeat the process to find a_2 using the model M_2

For AR Processes

Knowing that $\gamma_v(0) = 10, \gamma_v(\pm 1) = 0, \gamma_v(\pm 2) = 0, \gamma_v(\pm 3) = 3, \gamma_v(\tau) = 0 \; \forall \tau : |\tau| \geq 4$.
Consider the model class

$$M : v(t) = av(t-1) + bv(t-2) + cv(t-3) + \eta(t)$$

with $\eta \sim WN(0, \lambda^2)$

By using the prediction error minimization method, find the optimal estimates of the parameters a, b and c.

The predictor is equal to the starting equation but for the η component:

$$\hat{v}(t|t-1) = av(t-1) + bv(t-2) + cv(t-3)$$

The prediction error ε is always:

$$\varepsilon(t) = v(t) - \hat{v}(t|t-1)$$

Now let's compute J:

$$J = E[(\varepsilon(t))^2] =$$

$$E[v^2(t) + a^2v^2(t-1) + b^2v^2(t-1) + c^2v^2(t-3) +$$

$$-2av(t)v(t-1) - 2bv(t)v(t-2) - 2cv(t)v(t-3) +$$

$$2abv(t-1)v(t-2) + 2acv(t-1)v(t-3) + 2bcv(t-2)v(t-3)] =$$

$$\gamma_v(0) + a^2\gamma_v(0) + b^2\gamma_v(0) + c^2\gamma_v(0) - 2a\gamma_v(1) - 2b\gamma_v(2) - 2c\gamma_v(3) + 2ab\gamma_v(1) + 2ac\gamma_v(2) + 2bc\gamma_v(1)$$

$$J = \gamma_v(0)(1 + a^2 + b^2 + c^2) - 2c\gamma_v(3)$$

To minimize J we observe that we should impose $\hat{a} = \hat{b} = 0$

and, as always, let's derive wrt c:

$$\frac{\delta J}{\delta c} = 2c\gamma_v(0) - 2\gamma_v(3) \rightarrow \hat{c} = \frac{\gamma_v(3)}{\gamma_v(0)} = \frac{3}{10}$$

Parameter Estimation

Knowing Variance

MA

Usually asks which is the most suitable model class for representing the available data.
If you notice that $\gamma(\tau) = 0$ for $\tau > n$, where n is a finite number, you can say that the most suitable model class is a $MA(n)$.

Write the general equation for MA processes:

$$v(t) = e(t) + a_1 e(t-1) + a_2 e(t-2) + \ldots a_n e(t-n)$$

and use the known formulas for the covariances of MA processes to build a system of $n+1$ equations, then solve it.

Knowing Measurements

2. The measurements $y(t)$ of a time series for $t = -1,0,1,2,3,4$ are given in the table below.

t	-1	0	1	2	3	4
$y(t)$	0	1	-1	2	0	1

The aim is to identify, from the available data, a black-box model of the following class

$$m: y(t) = ay(t-1) + by(t-2) + e(t)$$

where $e(t) \approx WN(0, \lambda^2)$.

2.1. Compute the optimal estimate of a, b, and λ in such a way that the following cost function is minimized

$$J = \frac{1}{4}\sum_{t=1}^{4}(y(t) - \hat{y}(t/t-1))^2$$

dove $\hat{y}(t/t-1)$ optimal one-step predictor.

$A(z) = 1 - az^{-1} - bz^{-1}$ $C(z) = 1$

$\hat{y}(t|t-1) = a\,y(t-1) + b\,y(t-2)$

Therefore, defining $\varepsilon(t) = -\hat{y}(t|t-1) + y(t)$,

$\varepsilon(1) = -1 - a$
$\varepsilon(2) = 2 + a - b$
$\varepsilon(3) = b - 2a$
$\varepsilon(4) = 1 - 2b$

$\Rightarrow J = \frac{1}{4}\left\{(-1-a)^2 + (2+a-b)^2 + (b-2a)^2 + (1-2b)^2\right\}$

$$\begin{cases} \frac{\partial J}{\partial a} = 0 \\ \frac{\partial J}{\partial b} = 0 \end{cases}$$

WE OBTAIN $\Rightarrow$ THE SYSTEM:

$$\begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -3 \\ 4 \end{bmatrix} \Rightarrow \begin{matrix} \hat{a} = -\frac{2}{9} \\ \hat{b} = \frac{5}{9} \end{matrix}$$

$\hat{\lambda} = J(\hat{a}, \hat{b})$

Optimal Predictors via Long Division (only $e(t)$ case)

- take the process equation and put it in canonical form
- once you do it you'll have something in the form $y(t) = \frac{C(z)}{A(z)} e(t)$
- $C(z)$ on the left, $A(z)$ on the right. Do the long division:
  Let's for example compute the optimal 1-step predictor:
  Do the long division between $C(z)$ and $A(z)$. you'll obtain a quotient Q_1 and a remainder R_1, and such remainder can be called the 1-step remainder. R_1 and Q_1 will be used for the 1-step predictor in the following way:

  $$\hat{y}(t|t-1) = \frac{\hat{R}_1}{C(z)} y(t)$$

  where $\hat{R}_1$ is R_1 considering only the coefficients of z.
  So, if you want to compute the k-step predictor, just compute Q_K and R_k (which are the results of the k-th iteration of the long division) and use the generalize formula:

  $$\hat{y}(t|t-k) = \frac{\hat{R}_k}{C(z)} y(t)$$

  take care, if a z term is missing, for example $y(t) = 1 - z^{-2}$, the predictor of the missing z term is the same of the predictor of the next z term's corresponding step. so in this case the 1-step predictor is the same of the 2-steps predictor.